8 Lagrangian Formulation

While solving specific problems we have written the equation of motion of a particle in terms of the cartesian or the polar coordinates. Consider, for example, the motion of a particle in a central force-field. We studied this motion by using plane polar coordinates r and θ. The motion of a projectile is considered in the cartesian coordinate system. It is seen that a particular coordinate system chosen from the symmetry of the physical system helps to simplify the problem. Such a dependence on the coordinate system, however, is undesirable and we should be able to write the equations of motion without any specific reference to the coordinate system used. This is the approach in the Lagrangian formulation of Classical Mechanics. It has, as will be seen in this chapter, a number of advantages over the Newtonian formulation.

This approach may be compared to the use of vectors in describing physical quantities. When expressed through vectors, these equations do not depend on any specific coordinate system and the physical significance of the equation is also clearly seen. This is, however, a very limited comparison. The Lagrangian formulation is of a very general nature and, as we shall see presently, makes use of generalised coordinates and velocities which are independent of the coordinate system and may not be the usual spatial coordinates and velocities.

8.1 CONSTRAINTS

Consider the motion of a free particle. To describe this motion we can use three independent coordinates such as the cartesian coordinates x, y, z or the spherical polar coordinates r, θ, φ and so on. The particle is free to execute motion along any one axis independently with change in one coordinate only. The above statement is equivalent to saying that the particle has three degrees of freedom. The number of independent ways in which a mechanical system can move without violating any constraints which may be imposed on the system is called the number of degrees of freedom of that system. In other words, the number of degrees of freedom is the number of independent variables which must be simultaneously specified in order to describe the positions and velocities of all particles in the system for any motion which does not violate the constraints. Thus, for a system of N particles moving independently of each other, the number of degrees of freedom is obviously $3N$. For a particle constrained to move on a plane only two variables x, y or r, θ are sufficient to describe its motion and the particle is said to have two degrees of freedom. Thus, the constraint on the motion of the particle in a plane reduces the number of degrees of freedom by one. Further, let the particle be tied to one end of a rigid rod and be capable of rotating about the other end in the plane. The particle is then constrained to move along a circle of radius equal to the length of the rod. The motion of the particle can now be described in terms of a single variable θ and has only one degree of freedom.

When the motion of a system is restricted in some way, constraints are said to have been introduced. A bead sliding down a wire, a disc rolling down an inclined plane, etc. are some illustrations of constrained motion. Thus, when constraints are introduced into a system its number of degrees of freedom is reduced. Very often, we can express constraints in terms of certain equations. Thus, in the case of a rigid body, the constraint that the distance between any two particles of the body is a constant can be written as

$$|\mathbf{r}_i - \mathbf{r}_j|^2 = |\mathbf{r}_{ij}|^2 = (x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2$$
$$= \text{const} \tag{8.1}$$

where $\mathbf{r}_i$ and $\mathbf{r}_j$ are the position vectors of the ith and jth particles, respectively.

In the case of a simple pendulum moving in the xy-plane (Fig. 8.1), the two equations of the constraints are

$$z = 0 \text{ and } x^2 + y^2 = l^2 = \text{constant} \tag{8.2}$$

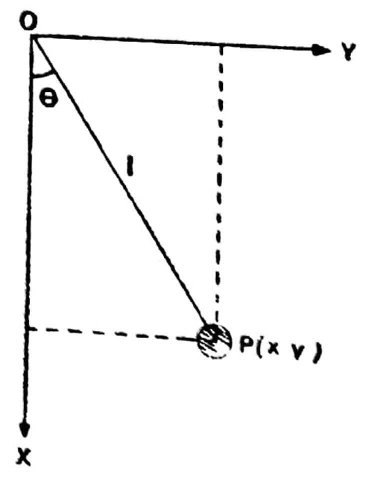

Fig. 8.1 Constraints on a simple pendulum.

It is clear from Fig. 8.1 that only one variable θ is sufficient to locate oscillating particle P

The equation of constraint in the case of a particle moving on or outside the surface of a sphere of radius a is

$$x^2 + y^2 + z^2 \geqslant a^2 \tag{8.3}$$

if the origin of the coordinate system coincides with the centre of the sphere (Fig. 8.2). The equality in expression (8.3) holds as long as the

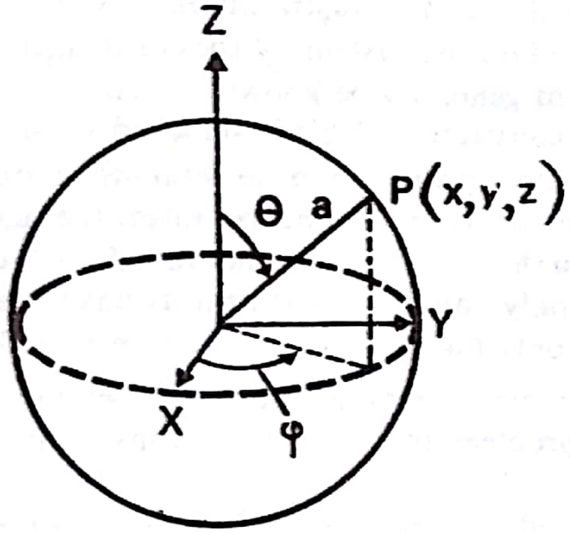

Fig. 8.2 Generalised coordinates of a particle constrained to move on the surface of a sphere

particle is in contact with the surface of the sphere. The inequality in expression (8.3) corresponds to a case when the particle leaves the surface, i.e. $r = \sqrt{x^2 + y^2 + z^2} > a$. Thus, a constraint is a restriction on the freedom of motion of a system of particles in the form of a condition which must be satisfied by their coordinates or by the allowed changes in their coordinates.

(a) Holonomic and Non-Holonomic Constraints

A holonomic constraint is one that may be expressed in the form of an equation relating the coordinates of the system and time. The general form of such equations for a system of N particles is

$$F_i[x_1, y_1, z_1, x_2, y_2, z_2, \ldots, x_N, y_N, z_N, t] = 0 \tag{8.4}$$

where $i = 1, 2, 3, \ldots, k$, and F_i is some function of the coordinates. In this equation, i denotes the ith constraint. It should be noted that all variables may not enter in the equation of constraints.

A non-holonomic constraint cannot be expressed in the type or form of equation (8.4). It may be in the form of an inequality as in expression (8.3).

If there are k equations of constraints, the number of degrees of freedom is reduced to $(3N - k)$. Hence, instead of $3N$ coordinates (x_i, y_i, z_i) where $i = 1, 2, \ldots, N$, we can assign the $(3N - k)$ independent variables $q_1, q_2, \ldots, q_{3N-k}$ to describe the system. Such variables are denoted by q_i, where $i = 1, 2, \ldots, (3N - k)$. These variables need not necessarily have the dimensions of length or angles, they may be even charges. Such variables are called the generalised coordinates. Thus, $q = \theta$ in the case of a simple pendulum, coordinates $q_1 = r$ and $q_2 = \theta$ in the case of the motion of a particle in a central force field are generalised coordinates. A set of independent coordinates $q_1, q_2, \ldots, q_{3N-k}$ is called a proper set of generalised coordinates.

It should be noted that the constraints on a system are the consequences of the forces exerted on the system by the constraining mechanism. This mechanism may not generally be known. Hence, it is necessary to eliminate the forces of constraint. This is achieved in the Lagrangian formulation by introducing a proper set of generalised coordinates. The choice of the generalised coordinates incorporates the constraints. This is possible only when the constraints are holonomic. We mostly come across systems in which only holonomic constraints have been introduced, hence we shall consider only the holonomic constraints in this book.

The non-holonomic constraints cannot be eliminated by any general method. Each problem involving such constraints needs to be solved individually.

It should be noted that the constraints need not always be written in the form of equations connecting the coordinates, but they can be expressed as well in terms of the velocities. For example, a disc of radius a rolling down an inclined plane along the line of greatest slope will have its constraint equation as follows (see Fig. 8.3). In this case

$$\frac{dl}{dt} = a\dot{\theta}$$

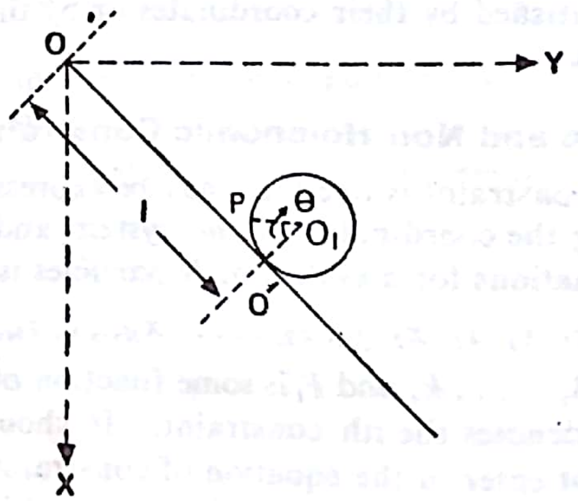

Fig. 8.3 Generalised coordinates of a disc rolling down an inclined plane without slipping

This can also be written as

$$dl = a\, d\theta$$

which after integration reduces to

$$l - a\theta = \text{const}$$

This is a holonomic constraint obtained from the relationship between the velocities.

(b) Scleronomous and Rheonomous Constraints

The constraints are also classified as

(i) scleronomous when the constraints are independent of time, and

(ii) rheonomous when the constraints are explicitly dependent on time.

Thus, the constraint in the case of a rigid body is scleronomous constraint while that on a bead on a rotating wire loop is rheonomous.

The various other illustrations mentioned above are those of scleronomous constraints. If, however, we construct a simple pendulum whose length changes with time, i.e., $l \equiv l(t)$, then the constraint expressed in equation (8.2) is time-dependent. Similarly if the radius of the sphere is changing with time, i.e., $r \equiv r(t)$, the constraint mentioned in expression (8.3) is time-dependent. These are, therefore, rheonomous constraints.

8.2 GENERALIZED COORDINATES

It is mentioned in article 8.1 that if on a system of particles there are k constraints, all the $3N$ coordinates of all the N particles in the system are not independent. Secondly, the forces of constraints are not always known because they may depend upon the motion itself. In such a case, we can introduce a proper set of variables, viz. $q_1, q_2, \ldots, q_n$, where $n = 1, 2, \ldots, (3N - k)$, to account for the forces of constraints. These $(3N - k)$ variables describe the system completely.

Let us suppose that the equations connecting $(x_1, y_1, z_1), (x_2, y_2, z_2), \ldots, (x_N, y_N, z_N)$ and $q_1, q_2, \ldots, q_n$ also depend upon time explicitly. The transformation equations are written as

$$\left.\begin{aligned} x_i &= x_i(q_1, q_2, \ldots, q_n, t) \equiv x_i(q_j, t) \\ y_i &= y_i(q_1, q_2, \ldots, q_n, t) \equiv y_i(q_j, t) \\ \text{and} \quad z_i &= z_i(q_1, q_2, \ldots, q_n, t) \equiv z_i(q_j, t) \end{aligned}\right\} \tag{8.5}$$

The three equations expressed in formulae (8.5) can be combined into a single vector equation

$$\mathbf{r}_i = \mathbf{r}_i(q_1, q_2, \ldots, q_n, t)$$

or

$$\mathbf{r}_i = \mathbf{r}_i(q_j, t) \tag{8.6}$$

where $i = 1, 2, \ldots, N$ and $j = 1, 2, \ldots, n$.

We shall, hereafter, use the vector form of equation (8.6) for further development of the theory.

As mentioned earlier, the new variable q_j may not necessarily have the dimensions of length. These may be angles such as θ and φ, as in the case of spherical polar coordinates or other convenient quantities. Let us consider some illustrations.

1. The motion of a simple pendulum (Fig. 8.1) oscillating in a vertical

plane can be described in terms of the cartesian coordinates x and y. But

$$x = l\cos\theta \quad \text{and} \quad y = l\sin\theta \tag{8.7}$$

Since l is a constant, the only variable involved is θ. It can be chosen as the generalised coordinate. Thus

$$q = \theta = \cos^{-1}\frac{x}{l} \quad \text{or} \quad \theta = \sin^{-1}\frac{y}{l}$$

2. The motion of a particle constrained to move on the surface of a sphere of radius a whose centre coincides with the origin of the coordinate system can be described in terms of the cartesian coordinates x, y and z. These coordinates satisfy the relation

$$x^2 + y^2 + z^2 = a^2$$

From this equation it is clear that only two coordinates are independent. Let us introduce $q_1 = \frac{x}{a}$ and $q_2 = \frac{y}{a}$. Then, we have

$$q_3 = \sqrt{\frac{a^2 - x^2 - y^2}{a^2}} = \sqrt{1 - q_1^2 - q_2^2} \tag{8.8}$$

Thus, q_3 is not an independent coordinate.

But, the coordinates chosen may not be the most convenient. This is because the problem has a spherical symmetry and hence the spherical polar coordinates can be used with greater advantage (Fig. 8.2). In these coordinates, $r = a =$ constant describes the sphere. Then, the convenient generalised coordinates will be $q_1 = \theta$ and $q_2 = \varphi$. These are given by

$$q_1 = \theta = \cos^{-1}\frac{z}{a}$$

and

$$q_2 = \varphi = \tan^{-1}\frac{y}{x} \tag{8.9}$$

3. Consider a disc rolling down an inclined plane without slipping. To specify the position of any point on the rim of the disc, we must give two coordinates. This is because the disc possesses both translational and rotational motion. We can conveniently use both the cartesian and the polar coordinates.

Any point P can be located by giving the position of the centre of disc O_1 or of the point of contact O' from reference point O and the angle made by the radial line with some fixed line. This line may be taken as a vertical line or the line joining the centre of the disc and the point of contact. The equations of the constraints for the axes chosen in Fig. 8.3 can be written as $z = 0$ and $l = \sqrt{x^2 + y^2}$, where l is the distance travelled by the disc down the plane.

If we locate the particle at point P by specifying the spherical polar coordinates with O' as the origin and $O'O_1$ as the polar axis, then $\varphi = 0$ and $O_1P =$ constant are the equations of the constraints. Thus, l and θ are the proper generalised coordinates for this system.

The generalised coordinates can be chosen in a variety of ways. We can very well describe the position of any point P on the disc by giving the vertical height of O' below the y-axis and the angle made by the position vector drawn from point O' to point P with the x- or y-axis. Similarly, we can choose the height of O' and the distance travelled by P in rolling down the plane, etc. All such alternative sets of independent coordinates are related to one another.

The inverse transformation equations, wherever they exist, can be written down as

$$q_j = q_j[x_1, y_1, z_1, x_2, y_2, z_2, \ldots, x_N, y_N, z_N, t] \tag{8.10}$$

or $$q_j = q_j(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N, t)$$

where $j = 1, 2, 3, \ldots, n$.

The time derivatives of the generalised coordinates, viz. $\dot{q} = \dfrac{dq}{dt}$, are called the generalised velocities. Once again, it should be borne in mind that the generalised velocities need not have the dimensions of velocities.

It is rather difficult to visualise the motion of a system of N particles having n degrees of freedom. This can, however, be done by imagining an n-dimensional space called the configuration space, denoted by q_j axes. The system at any instant is then represented by a point in the configuration space. As the system moves in physical space in time, its coordinates q_j change and the point in the configuration space also moves. Thus, the development of the system in a real 3-dimensional space can be represented by some curve in the configuration space. This is a convenient way of representing the real motion of a complex physical system by means of a 'trajectory' in the configuration space.

8.3 D'ALEMBERT'S PRINCIPLE

Consider a system described by n generalised coordinates q_j ($j = 1, 2, \ldots, n$). Suppose the system undergoes a certain displacement in the configuration space in such a way that it does not take any time and that it is consistent with the constraints on the system. Such displacements are called virtual because they do not represent actual displacements of the system. It should be noted that since there is no actual motion of the system, the work done by the forces of constraint in such a virtual displacement is zero. If a particle is constrained to move on the surface of a smooth sphere, then the force of constraint is equivalent to the reaction of the surface. In this case, the virtual displacement is taken at right angles to the direction of the force, i.e., along the surface, so that the work done by the force during the virtual displacement is zero.

Let the virtual displacement of the ith particle of the given system be $\delta\mathbf{r}_i$. If the given system is in equilibrium, the resultant force acting on the ith particle of the system must be zero, i.e. $\mathbf{F}_i = 0$. It is, then, obvious that virtual work $\mathbf{F}_i \cdot \delta\mathbf{r}_i = 0$ for the ith particle and hence it is also zero for all the particles of the system. Thus

$$dW = \sum_i \mathbf{F}_i \cdot \delta\mathbf{r}_i = 0 \tag{8.11}$$

The resultant force $\mathbf{F}_i$ acting on the ith particle is made up of two forces: $\mathbf{F}_i^a \rightarrow$ the applied force and $\mathbf{f}_i \rightarrow$ the force of constraint. Hence, we can write

$$\mathbf{F}_i = \mathbf{F}_i^a + \mathbf{f}_i \tag{8.12}$$

Equation (8.11) then becomes

$$\sum_i \mathbf{F}_i^a \cdot \delta\mathbf{r}_i + \sum_i \mathbf{f}_i \cdot \delta\mathbf{r}_i = 0 \tag{8.13}$$

Let us assume that the virtual work done by the forces of constraints is zero, i.e.

$$\sum_i \mathbf{f}_i \cdot \delta\mathbf{r}_i = 0 \tag{8.14}$$

In writing equation (8.14) we must imagine that virtual displacements $\delta\mathbf{r}_i$ are such that the total work done by the forces of constraints is zero. If this is not the case, some work will be done in the virtual displacements by the forces of constraint, the forces themselves being non-zero. Equation (8.14) will not hold if frictional forces are present. This is because the frictional forces act in a direction opposite to that of the displacement. We shall, therefore, exclude frictional forces from our considerations.

With this restriction, we arrive at the principle of virtual work which states that the *virtual work done by the applied forces acting on a system in equilibrium is zero, provided that no frictional forces are present.* Thus, we have

$$\sum_i \mathbf{F}_i^a \cdot \delta\mathbf{r}_i = 0 \tag{8.15}$$

The conditions under which the principle is applicable must be clearly borne in mind. It should also be noted that all the coordinates and, hence, the virtual displacements $\delta\mathbf{r}_i$ are not independent of each other. In fact, some of these must be connected by the equations of constraints. Hence, we cannot treat virtual displacements $\delta\mathbf{r}_i$ as completely arbitrary and equate their coefficients, viz. $\mathbf{F}_i^a$, to zero. For this purpose, we shall have to transform coordinates $\mathbf{r}_i$ to independent generalised coordinates q_j.

There is another point that needs consideration. Most of the systems we come across in mechanics are not in static equilibrium. Hence, the principle must be modified to include dynamic systems as well. This can be achieved by considering the equation of motion, viz.

$$\mathbf{F}_i = \dot{\mathbf{p}}_i$$

in the form

$$\mathbf{F}_i - \dot{\mathbf{p}}_i = 0 \tag{8.16}$$

Equation (8.16) states that the given dynamic system appears to be in equilibrium under the action of applied force $\mathbf{F}_i$ and an equal and opposite 'effective force' $\dot{\mathbf{p}}_i$. This effective force can be called the 'kinetic reaction'. When any particle is acted upon by an external force, the resultant of the applied force and the kinetic reaction is zero. A dynamic system considered in this manner appears to be static.

The principle of virtual work can be put into a useful form by the use of this approach. It is assumed that the *forces of constraint do no work.* Then, we can generalise equation (8.15) by the use of equation (8.16) to the form

$$\sum (\mathbf{F}_i - \dot{\mathbf{p}}_i)\cdot \delta \mathbf{r}_i = 0 \tag{8.17}$$

Equation (8.17) is the mathematical statement of D'Alembert's principle. In this equation all forces $\mathbf{F}_i$ are the applied forces. The forces of constraints do not appear in this equation for reasons mentioned above.

We now transform D'Alembert's principle into expressions containing independent generalised coordinates only. For this, we consider an infinitesimal virtual displacement $\delta \mathbf{r}_i$ at particular instant t. Then, from equation (8.6), we have

$$\delta \mathbf{r}_i = \sum_j \frac{\partial \mathbf{r}_i}{\partial q_j}\, \delta q_j \tag{8.18}$$

Variation with respect to time is absent in equation (8.18) because virtual displacement $\delta \mathbf{r}_i$ is assumed to take place at fixed instant t. Further the velocities are given by

$$\mathbf{v}_i = \dot{\mathbf{r}}_i = \sum_j \frac{\partial \mathbf{r}_i}{\partial q_j}\, \dot{q}_j + \frac{\partial \mathbf{r}_i}{\partial t} \tag{8.19}$$

where $\dot{q}_j = \dfrac{\partial q_j}{\partial t}$ are the generalised velocities. It should be noted that since generalised coordinates q_j need not have the dimensions of length, generalised velocities $\dot{q}_j$ need not have the dimensions of velocity.

The virtual work done by forces $\mathbf{F}_i$ in terms of virtual displacements $\delta \mathbf{r}_i$ is given by

$$\begin{aligned}
\delta W &= \sum_i \mathbf{F}_i \cdot \delta \mathbf{r}_i \\
&= \sum_j \left(\sum_i \mathbf{F}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} \right) \delta q_j \\
&= \sum_j Q_j\, \delta q_j
\end{aligned} \tag{8.20}$$

where $Q_j = \sum_i \mathbf{F}_i \cdot \dfrac{\partial \mathbf{r}_i}{\partial q_j}$ represents the jth component of the generalised force. We can express it as

$$\begin{aligned}
Q_j &= \sum_i \mathbf{F}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} \\
&= \sum_i \left(F_{ix} \frac{\partial x_i}{\partial q_j} + F_{iy} \frac{\partial y_i}{\partial q_j} + F_{iz} \frac{\partial z_i}{\partial q_j} \right)
\end{aligned} \tag{8.21}$$

It is obvious that just as generalised coordinates q_j need not have the dimensions of length, generalised force Q_j need not have the dimensions of force. But product $\sum_j Q_j\, \delta q_j$ must have the dimensions of work.

8.4 LAGRANGE'S EQUATIONS

We have already stated D'Alembert's principle in the form $\sum (\mathbf{F}_i - \dot{\mathbf{p}}_i)\cdot \delta \mathbf{r}_i = 0$ in equation (8.17). The first term, viz. $\sum_i \mathbf{F}_i \cdot \delta \mathbf{r}_i$, has already been proved equal to $\sum_j Q_j\, \delta q_j$ in equation (8.20).

Consider the second term on the right-hand side of this equation. We wish to express it in terms of the virtual displacements of the generalised coordinates. Thus, we can write

$$\sum_i \dot{\mathbf{p}}_i \cdot \delta\mathbf{r}_i = \sum_i \frac{d}{dt}(m_i \dot{\mathbf{r}}_i)\cdot\delta\mathbf{r}_i$$

$$= \sum_{ij} m_i \ddot{\mathbf{r}}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j}\,\delta q_j \tag{8.22}$$

The coefficient of δq_j on the right-hand side of equation (8.22) can be written as

$$\sum_i m_i \ddot{\mathbf{r}}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} = \sum_i \left[\frac{d}{dt}\left(m_i \dot{\mathbf{r}}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j}\right) - m_i \dot{\mathbf{r}}_i \cdot \frac{d}{dt}\left(\frac{\partial \mathbf{r}_i}{\partial q_j}\right)\right] \tag{8.23}$$

Now consider the part $\dfrac{d}{dt}\left(\dfrac{\partial \mathbf{r}_i}{\partial q_j}\right)$ of the last term in equation (8.23). In this we can interchange the order of differentiation by making use of equation (8.19). Thus

$$\frac{d}{dt}\left(\frac{\partial \mathbf{r}_i}{\partial q_j}\right) = \sum_k \frac{\partial^2 \mathbf{r}_i}{\partial q_j\,\partial q_k}\dot{q}_k + \frac{\partial^2 \mathbf{r}_i}{\partial q_j\,\partial t}$$

$$= \frac{\partial \dot{\mathbf{r}}_i}{\partial q_j} \tag{8.24}$$

by using equation (8.19) which gives the right-hand side as the expansion of $\dfrac{\partial \dot{\mathbf{r}}_i}{\partial q_j}$. Thus, equation (8.24) shows that the order of differentiation with respect to t and q_j can be interchanged.

Note that, in equation (8.24), a term dependent upon time is also included. Moreover, from equation (8.19), we have after differentiation with respect to $\dot{q}_j$

$$\frac{\partial \dot{\mathbf{r}}_i}{\partial \dot{q}_j} = \frac{\partial \mathbf{r}_i}{\partial q_j} \tag{8.25}$$

Substituting from equations (8.24) and (8.25) into equation (8.23), we get

$$\sum m_i \ddot{\mathbf{r}}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} = \sum_i \left[\frac{d}{dt}\left(m_i \dot{\mathbf{r}}_i \cdot \frac{\partial \dot{\mathbf{r}}_i}{\partial \dot{q}_j}\right) - m_i \dot{\mathbf{r}}_i \cdot \frac{\partial \dot{\mathbf{r}}_i}{\partial q_j}\right]$$

$$= \frac{d}{dt}\left[\frac{\partial}{\partial \dot{q}_j}\left(\sum_i \tfrac{1}{2} m_i |\dot{\mathbf{r}}_i|^2\right)\right] - \frac{\partial}{\partial q_j}\left[\left(\sum_i \tfrac{1}{2} m_i |\dot{\mathbf{r}}_i|^2\right)\right]$$

$$= \frac{d}{dt}\frac{\partial T}{\partial \dot{q}_j} - \frac{\partial T}{\partial q_j} \tag{8.26}$$

where T, the kinetic energy of the system, is given by

$$T = \sum_i \tfrac{1}{2} m_i |\dot{\mathbf{r}}_i|^2 \tag{8.27}$$

On combining equations (8.26), (8.22) and (8.20), D'Alembert's principle becomes

$$\sum_j \left[\left(\frac{d}{dt}\frac{\partial T}{\partial \dot{q}_j} - \frac{\partial T}{\partial q_j}\right) - Q_j\right]\delta q_j = 0 \tag{8.28}$$

Virtual displacements δq_j are all independent of one another. Hence, the set of n equations (8.28), where $j = 1, 2, 3, \ldots, n$, is true only if the coefficient of each displacement is zero. Thus, we get a set of n equations

$$\frac{d}{dt}\frac{\partial T}{\partial \dot{q}_j} - \frac{\partial T}{\partial q_j} = Q_j \tag{8.29}$$

The equations are valid in the case of conservative as well as non-conservative forces. These equations are called Lagrange's equations. However, this name is used mostly for equations of the systems whenever conservative forces are acting. In that case, the forces can be derivable from potential energy function V. Thus

$$F_{ix} = -\frac{\partial V}{\partial x_i},\ F_{iy} = -\frac{\partial V}{\partial y_i} \text{ and } F_{iz} = -\frac{\partial V}{\partial z_i} \tag{8.30}$$

In the vector notation, we can write

$$\mathbf{F}_i = -\nabla_i V$$

where
$$\nabla_i = \mathbf{i}\frac{\partial}{\partial x_i} + \mathbf{j}\frac{\partial}{\partial y_i} + \mathbf{k}\frac{\partial}{\partial z_i}$$

Potential energy function V is a function of $\mathbf{r}_i$ or q_j and is not a function of velocities $\dot{\mathbf{r}}_i$ or $\dot{q}_j$. Under these circumstances, the generalised forces are given by

$$\begin{aligned} Q_j &= \sum_i F_{ix}\frac{\partial x_i}{\partial q_j} + F_{iy}\frac{\partial y_i}{\partial q_j} + F_{iz}\frac{\partial z_i}{\partial q_j} \\ &= -\sum_i \left[\frac{\partial V}{\partial x_i}\frac{\partial x_i}{\partial q_j} + \frac{\partial V}{\partial y_i}\frac{\partial y_i}{\partial q_j} + \frac{\partial V}{\partial z_i}\frac{\partial z_i}{\partial q_j}\right] \\ &= -\frac{\partial V}{\partial q_j} \end{aligned} \tag{8.31}$$

Thus, the relation between potential energy function V and component Q_j of generalised conservative force is of the same form as given in equation (8.30). Moreover, $\frac{\partial V}{\partial \dot{q}_j} = 0$.

Hence, equation (8.29) can be written as

$$\frac{d}{dt}\frac{\partial (T-V)}{\partial \dot{q}_j} - \frac{\partial (T-V)}{\partial q_j} = 0 \tag{8.32}$$

Let the Lagrangian function L be defined by

$$\begin{aligned} L &= L(q_1, q_2, q_3, \ldots, q_n; \dot{q}_1, \dot{q}_2, \dot{q}_3, \ldots, \dot{q}_n) \\ &= T - V \end{aligned} \tag{8.33}$$

Then, equations (8.32) become

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_j} - \frac{\partial L}{\partial q_j} = 0 \tag{8.34}$$

which are known as Lagrange's equations of motion for a conservative holonomic system.

We have derived Lagrange's equations from D'Alembert's principle in which we have explicitly used Newton's equations of motion. Thus, the derivation of Lagrange's equations for a system is equivalent to Newton's equation of motion.

Importance of Lagrangian Formulation

As mentioned earlier, the Lagrangian formulation of mechanics is not a new theory; it is only an alternative and equivalent formulation. The Newtonian and Lagrangian equations of motion lead to second-order differential equations of motion of the system which, when solved, describe the nature of motion. However, the equations are obtained from different considerations that cause difference in the approach to the two formulations.

In the Newtonian approach, we are concerned with the applied forces acting on a system, the result of which is to accelerate the system. The forces are due to external agencies and they represent the cause of the acceleration produced in the body. The resulting accelerated motion is the effect of the force.

In the Lagrangian approach, we consider the kinetic and potential energies of the system. The concept of force does not enter into the discussion directly. This is the difference between the two formulations. The kinetic and the potential energies are scalar functions and are invariant under coordinate transformations. These transformations may as well include the transformations from the cartesian coordinates ($\mathbf{r}_i$) to the generalised coordinates (q_j). We have already seen that the generalised coordinates can be selected in many alternative ways and we can always find a set of generalised coordinates that makes the problem simpler to solve. In such transformations, it is easier to deal with scalar quantities than with vector quantities such as force, momentum, torque, etc., involved in the Newtonian formulation. Another difference is that in some problem it may not be possible to know all the forces acting on the system. This is particularly true if some constraints are introduced into the system. But, the expressions for the kinetic and potential energy may be given. Under these circumstances, the Lagrangian formulation is far more useful.

8.5 A GENERAL EXPRESSION FOR KINETIC ENERGY

In the derivation of Lagrange's equations, the kinetic energy of a system was expressed in terms of the cartesian components of the velocity. Thus

$$T = \sum_i \tfrac{1}{2} m_i \dot{\mathbf{r}}_i^2 = \sum_i \tfrac{1}{2} m_i \mathbf{v}_i^2 \tag{8.35}$$

We now wish to express the kinetic energy in terms of generalised coordinates q_j ($j = 1, 2, 3, \ldots, n$) for using it in Lagrange's equations.

Since $\mathbf{r}_i = \mathbf{r}_i(q_j, t)$, we can write

$$\dot{\mathbf{r}}_i = \sum_j \frac{\partial \mathbf{r}_i}{\partial q_j} \dot{q}_j + \frac{\partial \mathbf{r}_i}{\partial t} \tag{8.36}$$

Now, in order to get kinetic energy T, we have to square expression (8.36). This will give three types of terms: the term involving square of the velocity, the term involving the velocity and the term independent of the velocity. Squaring equation (8.36), we have

$$\dot{r}_i^2 = \sum_{jk} \frac{\partial \mathbf{r}_i}{\partial q_j} \cdot \frac{\partial \mathbf{r}_i}{\partial q_k} \dot{q}_j \dot{q}_k + 2 \sum_j \frac{\partial \mathbf{r}_i}{\partial q_j} \cdot \frac{\partial \mathbf{r}_i}{\partial t} \dot{q}_j + \left(\frac{\partial \mathbf{r}_i}{\partial t}\right)^2 \tag{8.37}$$

Substituting this value of $\dot{r}_i^2$ in equation (8.35), we get

$$T = \sum_{jk} a_{jk} \dot{q}_j \dot{q}_k + \sum_j b_j \dot{q}_j + c \tag{8.38}$$

where

$$\left.\begin{aligned} a_{jk} &= \tfrac{1}{2} \sum_i m_i \frac{\partial \mathbf{r}_i}{\partial q_j} \cdot \frac{\partial \mathbf{r}_i}{\partial q_k} \\ b_j &= \sum_i m_i \frac{\partial \mathbf{r}_i}{\partial q_j} \cdot \frac{\partial \mathbf{r}_i}{\partial t} \\ \text{and} \qquad c &= \tfrac{1}{2} \sum_i m_i \left(\frac{\partial \mathbf{r}_i}{\partial t}\right)^2 \end{aligned}\right\} \tag{8.39}$$

The case when transformation equations (8.6) are independent of time is of particular interest. The constraints are then scleronomous and the partial derivatives with respect to time vanish. Hence, $b_j = 0$ and $c = 0$. In this case, the kinetic energy of the system is a homogeneous quadratic function of the generalised velocities. Thus

$$T = \sum_{jk} a_{jk} \dot{q}_j \dot{q}_k \tag{8.40}$$

Let us now form the sum $\sum_l \dot{q}_l \dfrac{\partial T}{\partial \dot{q}_l}$. From equation (8.40), we have

$$\frac{\partial T}{\partial \dot{q}_l} = \sum_i a_{il} \dot{q}_i + \sum_j a_{lj} \dot{q}_j$$

While carrying out the differentiation, we have used $\dfrac{\partial \dot{q}_i}{\partial \dot{q}_l} = \delta_{il}$, since $\dot{q}_j$ are a set of independent velocities. Here, the suffixes i and j are dummy indices as these can be replaced by other suffixes without changing the value of the sum. The sum $\sum \dot{q}_l \dfrac{\partial T}{\partial \dot{q}_l}$ is then written as

$$\sum_l \dot{q}_l \frac{\partial T}{\partial \dot{q}_l} = \sum_{il} a_{il} \dot{q}_i \dot{q}_l + \sum_{jl} a_{lj} \dot{q}_l \dot{q}_j$$

Since, all the suffixes are dummy and are symmetrical in i and j, the two sums on the right-hand side are identical. Hence

$$\sum_l \dot{q}_l \frac{\partial T}{\partial \dot{q}_l} = 2 \sum_{il} a_{il} \dot{q}_i \dot{q}_l = 2T \tag{8.41}$$

The result of the equation follows immediately from Euler's theorem which states that if $f(x_i)$ is a homogeneous function of the nth degree of a set of variables x_i, then

$$\sum x_i \frac{\partial f}{\partial x_i} = nf \tag{8.42}$$

Thus, in obtaining Lagrange's equation of motion of a system, we have to form the Lagrangian L of that system, which is the difference between the kinetic and potential energy of the system. Once this is done, the remaining procedure is mechanical. But, it is very important to get a correct expression for kinetic energy T. It is always better for a beginner to start from the expression for T in the cartesian coordinates and then substitute proper values of $\dot{x}_i$, $\dot{y}_i$ and $\dot{z}_i$ in it from the transformation equations.

The procedure is generally lengthy and tedious, and it is sometimes found convenient to write the velocity directly in terms of the generalised coordinates and obtain the expression for the kinetic energy.

Consider an illustration of a double pendulum which is a system of two pendulums, the second pendulum of length l_2 being suspended from the bob of the first pendulum of length l_1 (Fig. 8.4). Let us assume that the pendulums move in the xy plane only. The generalised coordinates in this case are $q_1 = \theta_1$ and $q_2 = \theta_2$ which are the angular displacements of the first and second pendulums from the vertical respectively. It is difficult to find an expression for kinetic energy T directly in terms of $\dot{\theta}_1$ and $\dot{\theta}_2$. We, therefore, write the transformation equations of the two bobs whose coordinates are (x_1, y_1) and (x_2, y_2) respectively. From Fig. 8.4, we have

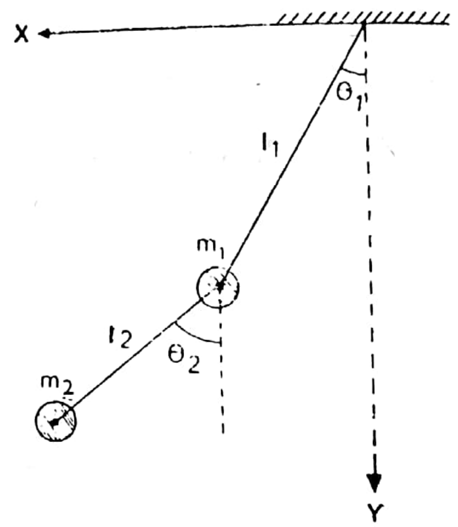

Fig. 8.4 Double pendulum

$$\left.\begin{aligned} x_1 &= l_1 \sin \theta_1 \\ y_1 &= l_1 \cos \theta_1 \\ x_2 &= l_1 \sin \theta_1 + l_2 \sin \theta_2 \\ \text{and} \quad y_2 &= l_1 \cos \theta_1 + l_2 \cos \theta_2 \end{aligned}\right\} \tag{8.43}$$

Differentiating equations (8.43) with respect to time, we get

$$\left.\begin{aligned} \dot{x}_1 &= l_1 \cos \theta_1 \dot{\theta}_1 \\ \dot{y}_1 &= -l_1 \sin \theta_1 \dot{\theta}_1 \\ \dot{x}_2 &= l_1 \cos \theta_1 \dot{\theta}_1 + l_2 \cos \theta_2 \dot{\theta}_2 \\ \text{and} \quad \dot{y}_2 &= -l_1 \sin \theta_1 \dot{\theta}_1 - l_2 \cos \theta_2 \dot{\theta}_2 \end{aligned}\right\} \tag{8.44}$$

Now, kinetic energy T is given by

$$T = \tfrac{1}{2} m_1 (\dot{x}_1^2 + \dot{y}_1^2) + \tfrac{1}{2} m_2 (\dot{x}_2^2 + \dot{y}_2^2)$$

Substituting the values of $\dot{x}_1, \dot{y}_1, \ldots,$ etc. and simplifying, we get

$$T = \tfrac{1}{2}(m_1 + m_2) l_1^2 \dot{\theta}_1^2 + \tfrac{1}{2} m_2 l_2^2 \dot{\theta}_2^2 + m_2 l_1 l_2 \cos (\theta_1 - \theta_2) \dot{\theta}_1 \dot{\theta}_2 \tag{8.45}$$

This expression can also be obtained by using equation (8.40). Thus

$$T = \sum_{jk} a_{jk} \dot{q}_j \dot{q}_k$$

$$= a_{\theta_1\theta_1} \dot{\theta}_1^2 + (a_{\theta_1\theta_2} + a_{\theta_2\theta_1}) \dot{\theta}_1 \dot{\theta}_2 + a_{\theta_2\theta_2} \dot{\theta}_2^2 \tag{8.46}$$

where

$$a_{\theta_1\theta_1} = \tfrac{1}{2}m_1\left[\left(\frac{\partial x_1}{\partial\theta_1}\right)^2 + \left(\frac{\partial y_1}{\partial\theta_1}\right)^2\right] + \tfrac{1}{2}m_2\left[\left(\frac{\partial x_2}{\partial\theta_1}\right)^2 + \left(\frac{\partial y_2}{\partial\theta_1}\right)^2\right]$$

$$a_{\theta_1\theta_2} = a_{\theta_2\theta_1} = \tfrac{1}{2}m_1\left[\frac{\partial x_1}{\partial\theta_1}\frac{\partial x_1}{\partial\theta_2} + \frac{\partial y_1}{\partial\theta_1}\frac{\partial y_1}{\partial\theta_2}\right] + \tfrac{1}{2}m_2\left[\frac{\partial x_2}{\partial\theta_1}\frac{\partial x_2}{\partial\theta_2} + \frac{\partial y_2}{\partial\theta_1}\frac{\partial y_2}{\partial\theta_2}\right]$$

and $$a_{\theta_2\theta_2} = \tfrac{1}{2}m_1\left[\left(\frac{\partial x_1}{\partial\theta_2}\right)^2 + \left(\frac{\partial y_1}{\partial\theta_2}\right)^2\right] + \tfrac{1}{2}m_2\left[\left(\frac{\partial x_2}{\partial\theta_2}\right)^2 + \left(\frac{\partial y_2}{\partial\theta_2}\right)^2\right]$$

Substituting the values of the various derivatives and simplifying, we get

$$a_{\theta_1\theta_1} = \tfrac{1}{2}(m_1 + m_2)l_1^2$$

$$a_{\theta_1\theta_2} = a_{\theta_2\theta_1} = \tfrac{1}{2}m_2 l_1 l_2 \cos(\theta_1 - \theta_2) \tag{8.47}$$

and $$a_{\theta_2\theta_2} = \tfrac{1}{2}m_2 l_2^2$$

Substituting these values of a's in the expression for T, we get the same expression for the kinetic energy as before.

8.6 SYMMETRIES AND THE LAWS OF CONSERVATION

So far, we have discussed Lagrange's dynamical equations of a system. If the system under consideration has n degrees of freedom, we get n second-order differential equations. The solution of each equation will need the evaluation of a double integral and will, therefore, involve two constants—initial position q_0 and initial velocity $\dot{q}_0$. Naturally, the solutions of n differential equations will involve $2n$ constants—n values of initial q's and $\dot{q}$'s.

In many problems, the solution cannot be obtained in terms of known functions. Moreover, sometimes solutions of the type $q_j = q_j(t)$ are of no interest to us. For example, when we study the motion of systems consisting of atoms and molecules, we are only interested in the evaluation of quantities such as energies and angular momenta. However, information regarding the physical nature of the motion of the system can often be extracted without integrating the equations of motion.

On considering the symmetries of the system, one can immediately obtain first integrals of the equations of motion. The first integrals are constants of motion. These are the first-order differential equations of the type

$$f(q_1, q_2, \ldots, q_n; \dot{q}_1, \dot{q}_2, \ldots, \dot{q}_n, t) = \text{const} \tag{8.48}$$

It is obvious that the first integral contains the first derivative of q's. The first integrals reveal a lot of information regarding the system under consideration. In fact, the conserved quantities discussed in Chapter 3 are first integrals of motion.

Consider a system of particles in a conservative force-field. Then, potential energy V depends only upon the position and we have

$$\frac{\partial L}{\partial \dot{x}_i} = \frac{\partial(T-V)}{\partial \dot{x}_i} = \frac{\partial}{\partial \dot{x}_i}\sum_j \tfrac{1}{2}m_j(\dot{x}_j^2 + \dot{y}_j^2 + \dot{z}_j^2)$$

$$= m_i\dot{x}_i = p_{x_i}$$

where p_{x_i} is the x-component of the linear momentum of the ith particle of the system. We can generalise this result and define the generalised momentum by the formula

$$p_j = \frac{\partial L}{\partial \dot{q}_j} \tag{8.49}$$

Quantity p_j is very often called the *canonical* or the *conjugate* momentum. If q_j represents translation (i.e. linear displacement), then p_j represents linear momentum (i.e., mass multiplied by velocity). But, if q_j represents an angle, p_j represents the angular momentum. The generalised definition of momentum allows us to consider non-mechanical systems as well. If we consider a charged particle moving in an electromagnetic field, then the Lagrangian is defined as

$$L = \tfrac{1}{2}m|\dot{\mathbf{r}}|^2 - q\Phi + q\mathbf{A}\cdot\dot{\mathbf{r}}$$

where Φ is a scalar potential and $\mathbf{A}$ is a vector potential. Differentiating with respect to $\dot{x}$, we get

$$p_x = m\dot{x} + qA_x$$

as the canonical momentum to x. It is not the usual kinetic momentum $m\dot{x}$ but has a contribution of qA_x from the electromagnetic field.

It should be noted that with this definition, Lagrange's equations for a conservative system assume a simple form similar to Newton's equations of motion. These are

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_j} = \dot{p}_j = \frac{\partial L}{\partial q_j} = -\frac{\partial V}{\partial q_j} = Q_j$$

or

$$\dot{p}_j = Q_j \tag{8.50}$$

8.7 CYCLIC OR IGNORABLE COORDINATES

The Lagrangian L is written as a function of q_j and $\dot{q}_j$. If any one coordinate, say q_k, is absent in the expression for the Lagrangian L, then the partial derivative of L with respect to q_k will vanish, i.e.

$$\frac{\partial L}{\partial q_k} = 0$$

Thus, the equation of motion corresponding to variable q_k becomes

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_k}\right) = 0 \tag{8.51}$$

Integrating equation (8.51), we get

$$\frac{\partial L}{\partial \dot{q}_k} = p_k = \text{const}$$

Thus, whenever coordinate q_k does not appear explicitly in the Lagrangian function L, corresponding linear momentum p_k is a constant of the motion, or in other words, it is the first integral of the motion. Such a coordinate q_k is said to be *cyclic* or *ignorable*.

Thus, we are led to the conservation of momentum—linear or angular—conjugate to the cyclic coordinate. The laws of conservation discussed in

Chapter 3 are, in general, contained in the conservation theorem relating to the cyclic coordinates.

There should be a deeper relationship between a cyclic coordinate and the physical nature of the motion of the system. When generalised translation coordinate q_k is cyclic, i.e., absent in the Lagrangian function, it means that the system can be translated along q_k without any effect on the Lagrangian. The Lagrangian function L is unchanged under this translation and the equation of motion remains the same. Thus, the system is invariant under translation or the system is said to have translational symmetry. This immediately leads us to the conservation of conjugate momentum. To illustrate this, consider the motion of two bodies under the action of internal forces. As no external force is acting on the system, the translation of the whole system will not affect the internal forces or the velocities. Thus, kinetic energy T, potential energy V and hence, the Lagrangian function L are unaffected. If $\mathbf{R}(X, Y, Z)$ represents the position vector of the centre of mass of the system, then

$$\frac{\partial L}{\partial X} = \frac{\partial L}{\partial Y} = \frac{\partial L}{\partial Z} = 0$$

Hence, the linear momentum of the centre of mass of the system, i.e., $M\dot{\mathbf{R}}$ is conserved. This result was obtained in Chapter 3. Similar considerations can be applied to a system having rotational symmetry. If the rotation of the system about a particular axis leaves the system unaffected, then the corresponding coordinate is cyclic and the component of the angular momentum in that direction is conserved.

We have seen that the conservation of the linear and the angular momentum of a system corresponds to its translational and rotational symmetries. This approach involving the considerations of symmetries is very fundamental and is helpful in many modern fields of physics.

It, therefore, appears that the conservation of energy of a system must also correspond to some symmetry of the system. This symmetry is with respect to the translation of the system along time. If the Lagrangian function L of a system does not explicitly depend upon time, then the translation of the system along time t leaves it unchanged. Then, we can write

$$\frac{\partial L}{\partial t} = 0 \tag{8.52}$$

Hence, the total time derivative of $L \equiv L(q_j, \dot{q}_j)$ is given by

$$\frac{dL}{dt} = \sum_j \frac{\partial L}{\partial q_j} \dot{q}_j + \sum_j \frac{\partial L}{\partial \dot{q}_j} \ddot{q}_j \tag{8.53}$$

It should be noted that the partial time derivative does not occur in equation (8.53) where we have used equation (8.52). Using Lagrange's equation, we can rewrite equation (8.53) as

$$\frac{dL}{dt} = \sum_j \dot{q}_j \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right) + \sum_j \frac{\partial L}{\partial \dot{q}_j} \ddot{q}_j$$

Hence,
$$\frac{dL}{dt} = \sum_j \frac{d}{dt}\left(\dot{q}_j \frac{\partial L}{\partial \dot{q}_j}\right)$$

or
$$\frac{d}{dt}\left(L - \sum_j \dot{q}_j \frac{\partial L}{\partial \dot{q}_j}\right) = 0 \tag{8.54}$$

Thus, the quantity in the bracket on the left-hand side of equation (8.54) is constant in time. It is denoted by $-H$. Hence, we have

$$H = \sum_j \dot{q}_j \frac{\partial L}{\partial \dot{q}_j} - L$$
$$= \sum_j \dot{q}_j p_j - L \tag{8.55}$$

If, potential energy V does not depend upon velocity or time, i.e. $V \equiv V(q_j)$ only, we can write

$$p_j = \frac{\partial L}{\partial \dot{q}_j} = \frac{\partial}{\partial \dot{q}_j}(T - V) = \frac{\partial T}{\partial \dot{q}_j}, \text{ simply.}$$

With this, the first term in equation (8.55) is $2T$ from equation (8.41). The transformation equations connecting $\mathbf{r}_i$ and q_j do not contain time explicitly, i.e., the constraints and the forces are independent of time and the kinetic energy is a quadratic function of velocities. Hence, equation (8.55) becomes

$$H = 2T - L = 2T - (T - V) = T + V = E \tag{8.56}$$

where E is the total energy of the system and is a constant of the motion.

Function H introduced in equation (8.55) is called the Hamiltonian of the system. It is equal to the total energy of the system only when the kinetic energy of the system is a homogeneous quadratic function of the velocities and the potential energy is independent of the velocities and time. The statements that kinetic energy T is a homogeneous quadratic function of velocities, or that the transformation equation $\mathbf{r}_i = \mathbf{r}_i(q_j)$ and the constraints are independent of time, are equivalent.

Thus, we have obtained the law of conservation of energy for a conservative system having time-independent constraints. This last restriction was not necessary for the law in Chapter 3.

In Chapter 3, the changes in energy (kinetic and potential) were due to the work done by all the forces including the forces of constraint. In the Lagrangian formulation, the forces of constraint were removed and the potential energy includes the work done by the applied forces only. When the constraints depend upon time, they also do work during the displacement which is included in potential energy V and hence total energy E will not be equal to the Hamiltonian H. When the constraints are time-independent, it will be recalled that the virtual work done by the forces of constraint is zero. Further, in this case, the virtual displacement which was taken at fixed instant t can now be taken in an interval of time δt and the difference between the real and the virtual displacements disappears. In that case, total energy E will be equal to the Hamiltonian H.

It should be noted that if the constraints are time dependent or the transformation equations $\mathbf{r}_i = \mathbf{r}_i(q_j, t)$ contain time explicitly, $H \neq E$; but still total energy E is conserved.

1. *Illustrations*: Consider the motion of a particle due to force $\mathbf{F}$ having components F_x, F_y and F_z along the three axes of the cartesian coordinates. Kinetic energy T of the particle is given by

$$T = \tfrac{1}{2}(\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$$

Therefore

$$\frac{\partial T}{\partial x} = \frac{\partial T}{\partial y} = \frac{\partial T}{\partial z} = 0$$

and
$$\frac{\partial T}{\partial \dot{x}} = m\dot{x}, \quad \frac{\partial T}{\partial \dot{y}} = m\dot{y} \text{ and } \frac{\partial T}{\partial \dot{z}} = m\dot{z}$$

Hence, the equations of motion are

$$\frac{d}{dt}\frac{\partial T}{\partial \dot{x}} = \frac{d}{dt}(m\dot{x}) = F_x \ldots \text{etc.}$$

These are nothing but Newton's equations of motion.

2. Consider now a bead sliding along a uniformly rotating wire in a force-free space. The transformation equations relating the cartesian and polar coordinates of the bead are

$$x = r\cos\theta = r\cos\omega t$$

and
$$y = r\sin\theta = r\sin\omega t$$

where ω is the constant angular velocity of the wire.

Kinetic energy T is given by

$$\begin{aligned} T &= \tfrac{1}{2}m(\dot{x}^2 + \dot{y}^2) \\ &= \tfrac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) \\ &= \tfrac{1}{2}m(\dot{r}^2 + r^2\omega^2) \end{aligned}$$

Thus, we find that T is not a homogeneous quadratic function of velocities only. The equation of motion can be written as

$$m\ddot{r} = mr\omega^2$$

or
$$\ddot{r} = r\omega^2$$

This is the familiar expression of the centripetal acceleration. This method used to obtain the equation of motion does not at all involve the force of constraint that keeps the bead on the wire.

3. *Atwood's machine*: Atwood's machine (Fig. 8.5) is an illustration of a simple mechanical system with a holonomic constraint. It consists of two masses m_1 and m_2 tied together by means of a light inextensible cord of length l. The cord passes round a light frictionless pulley and the two masses hang on the two sides of the pulley. We find from Fig. 8.5 that only one variable x is independent, since length l of the

string is constant. The kinetic and the potential energies of the system are given by

$$T = \tfrac{1}{2}m_1\dot{x}^2 + \tfrac{1}{2}m_2(\dot{l} - \dot{x})^2$$

and
$$V = -m_1gx - m_2g(l - x)$$

Hence, the Lagrangian function is given by

$$L = \tfrac{1}{2}(m_1 + m_2)\dot{x}^2 + m_1gx + m_2g(l - x)$$

since $\dot{l} = 0$.

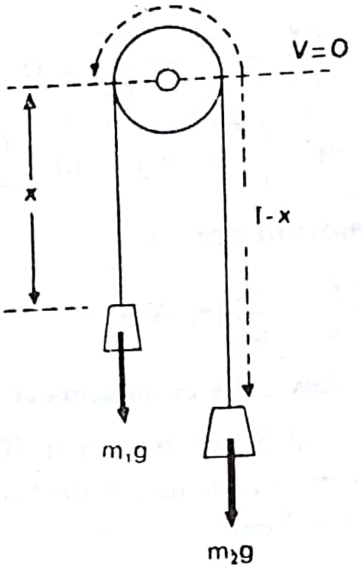

Fig. 8.5 Atwood's machine

The equation of motion of the system is, therefore, given by

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x} = (m_1 + m_2)\ddot{x} - (m_1 - m_2)g = 0$$

or
$$\ddot{x} = \frac{m_1 - m_2}{m_1 + m_2}g$$

Integrating this equation twice, we get

$$x = x_0 + v_0t + \tfrac{1}{2}\left(\frac{m_1 - m_2}{m_1 + m_2}\right)gt^2$$

which is a familiar equation. In this problem, the force of constraint—the tension in the cord—does not appear in the above equations and hence it cannot be found out directly by this method.

4. *Spherical pendulum*: A spherical pendulum is essentially a simple pendulum which is free to move through the entire space about the point of suspension (Fig. 8.6). The bob of such a pendulum moves on the surface of a sphere whose radius is equal to the length of the pendulum. It is obvious that spherical polar coordinates are most suitable to locate the position of the bob of the pendulum. Moreover, since l is a constant, only θ and φ need be specified.

The velocity of the bob is given by

$$\mathbf{v} = l\dot{\theta}\hat{\mathbf{e}}_\theta + l\sin\theta\dot{\varphi}\hat{\mathbf{e}}_\varphi$$

Hence, the kinetic energy of the pendulum is given by

$$T = \tfrac{1}{2}mv^2$$
$$= \tfrac{1}{2}m(l^2\dot{\theta}^2 + l^2 \sin^2 \theta\dot{\varphi}^2)$$

Potential energy V due to gravity is $V = -mgl \cos \theta$; it is measured with reference to the horizontal plane $z = 0$. Hence, the Lagrangian function L is given by

$$L = T - V = \tfrac{1}{2}ml^2\dot{\theta}^2 + \tfrac{1}{2}ml^2 \sin^2 \theta\dot{\varphi}^2 + mgl \cos \theta$$

Then, the Lagrangian equations corresponding to coordinates θ and φ are given by

$$\frac{d}{dt}(ml^2\dot{\theta}) - ml^2 \sin \theta \cos \theta\dot{\varphi}^2 - mgl \sin \theta = 0$$

and
$$\frac{d}{dt}(ml^2 \sin^2 \theta\dot{\varphi}) = 0$$

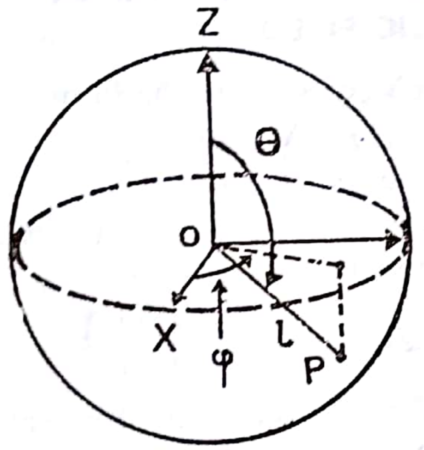

Fig. 8.6 Spherical pendulum

Variable φ is ignorable and hence the corresponding momentum is conserved. Thus

$$p_\varphi = ml^2 \sin^2 \theta\dot{\varphi} = \text{const}$$

Substituting the value of φ in terms of p_φ, we get the one-dimensional equation of motion in θ, viz.

$$ml^2\ddot{\theta} = mgl \sin \theta + \frac{p_\varphi^2 \cos \theta}{ml^2 \sin^3 \theta}$$

We can find the total energy of the pendulum as follows:

Since $\dfrac{\partial L}{\partial t} = 0$, total energy E is given by equation (8.55) as

$$E = \dot{\theta}\frac{\partial L}{\partial \dot{\theta}} + \dot{\varphi}\frac{\partial L}{\partial \dot{\varphi}} - L$$
$$= \tfrac{1}{2}ml^2\dot{\theta}^2 + \tfrac{1}{2}ml^2 \sin^2 \theta\dot{\varphi}^2 + mgl \cos \theta$$

and is a constant of the motion. This can easily be verified by taking total time derivative of E and showing it to be equal to zero.

The expression for the total energy can be rewritten after substituting for $\dot{\varphi}$ as

$$E = \tfrac{1}{2}ml^2\dot{\theta}^2 + \frac{p_\varphi^2}{2ml^2 \sin^2 \theta} + mgl \cos \theta$$

$$= T + V_e$$

where $V_e = mgl \cos \theta + \dfrac{p_\varphi^2}{2ml^2 \sin^2 \theta}$ is the effective potential energy and it depends only upon θ. In case the pendulum is restricted to move only in one plane, say the $\varphi = 0$ plane, the equation of motion reduces to

$$\ddot{\theta} = \frac{g}{l} \sin \theta$$

which is the familiar equation occurring in the case of a simple pendulum with the difference that θ here is measured with respect to the vertically upward direction.

8.8 VELOCITY-DEPENDENT POTENTIAL OF ELECTROMAGNETIC FIELD

We can obtain Lagrange's equation in the form

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right) - \frac{\partial L}{\partial q_j} = 0$$

even if the system is not conservative in the usual sense provided that the generalised forces are expressible in the form

$$Q_j = -\frac{\partial U}{\partial q_j} + \frac{d}{dt}\left(\frac{\partial U}{\partial \dot{q}_j}\right) \tag{8.57}$$

where $U(q_j, \dot{q}_j)$ may be called the 'generalised potential' or the 'velocity-dependent potential'. We come across such a potential when we consider the electromagnetic forces acting on moving charges.

Maxwell's equations are stated as

$$\left.\begin{aligned} \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} &= 0, \qquad \nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \\ \nabla \times \mathbf{B} &= \mu_0 j + \mu_0\epsilon_0 \frac{\partial \mathbf{E}}{\partial t}, \qquad \nabla \cdot \mathbf{B} = 0 \end{aligned}\right\} \tag{8.58}$$

Quantity $\epsilon_0\mathbf{E} = \mathbf{D}$ is called the electric displacement while the magnetic induction is $\mathbf{B} = \mu_0\mathbf{H}$.

The force on a charged particle having charge q and moving in an electromagnetic field is given by

$$\mathbf{F} = q[\mathbf{E} + \mathbf{v} \times \mathbf{B}] \tag{8.59}$$

and is known as the 'Lorentz force'.

From equation (8.58), we observe that $\nabla \times \mathbf{E} \neq 0$. Hence, we cannot express $\mathbf{E}$ as a gradient of a scalar function. But, $\nabla \cdot \mathbf{B} = 0$ suggests that we can express $\mathbf{B}$ as a curl of vector $\mathbf{A}$. Thus

$$\mathbf{B} = \nabla \times \mathbf{A} \tag{8.60}$$

where **A** is called a magnetic vector potential. With this substitution for **B**, the equation

$$\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0$$

becomes

$$\nabla \times \mathbf{E} + \frac{\partial}{\partial t}(\nabla \times \mathbf{A}) = 0$$

or

$$\nabla \times \left[\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t}\right] = 0 \tag{8.61}$$

Hence, we can write

$$\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} = -\nabla \Phi$$

or

$$\mathbf{E} = -\nabla \Phi - \frac{\partial \mathbf{A}}{\partial t} \tag{8.62}$$

The Lorentz force can now be written as

$$\mathbf{F} = q\left[-\nabla \Phi - \frac{\partial \mathbf{A}}{\partial t} + \mathbf{v} \times \nabla \times \mathbf{A}\right] \tag{8.63a}$$

Consider the x-components of the various terms of equation (8.63a). We have

$$[\nabla \Phi]_x = \frac{\partial \Phi}{\partial x}$$

$$[\mathbf{v} \times \nabla \times \mathbf{A}]_x = v_y\left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right] - v_z\left[\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right] \tag{8.63b}$$

$$= v_y \frac{\partial A_y}{\partial x} + v_z \frac{\partial A_z}{\partial x} - v_y \frac{\partial A_x}{\partial y} - v_z \frac{\partial A_x}{\partial z}$$

Now, we add and subtract term $v_x \dfrac{\partial A_x}{\partial x}$ to the right-hand side of this expression. Then, we get

$$[\mathbf{v} \times \nabla \times \mathbf{A}]_x = v_x \frac{\partial A_x}{\partial x} + v_y \frac{\partial A_y}{\partial x} + v_z \frac{\partial A_z}{\partial x} - v_x \frac{\partial A_x}{\partial x} - v_y \frac{\partial A_x}{\partial y} - v_z \frac{\partial A_x}{\partial z} \tag{8.64}$$

Further, the total time derivative of $A_x \equiv A_x(x, y, z, t)$ is

$$\frac{dA_x}{dt} = \frac{\partial A_x}{\partial t} + \left(\frac{\partial A_x}{\partial x}\frac{dx}{dt} + \frac{\partial A_y}{\partial y}\frac{dy}{dt} + \frac{\partial A_z}{\partial z}\frac{dz}{dt}\right)$$

or

$$\frac{dA_x}{dt} = \frac{\partial A_x}{\partial t} + \left(v_x \frac{\partial A_x}{\partial x} + v_y \frac{\partial A_y}{\partial y} + v_z \frac{\partial A_z}{\partial z}\right) \tag{8.65}$$

Hence, we can write

$$[\mathbf{v} \times \nabla \times \mathbf{A}]_x = \frac{\partial}{\partial x}(\mathbf{v} \cdot \mathbf{A}) - \frac{dA_x}{dt} + \frac{\partial A_x}{\partial t}$$

With these substitutions, the x-component of the Lorentz force is given by

$$F_x = q\left[-\frac{\partial}{\partial x}\{\Phi - \mathbf{v} \cdot \mathbf{A}\} - \frac{d}{dt}\left\{\frac{\partial}{\partial v_x}(\mathbf{A} \cdot \mathbf{v})\right\}\right] \tag{8.66}$$

Now, scalar potential Φ is independent of the velocity; hence we can say that

$$F_x = -\frac{\partial U}{\partial x} + \frac{d}{dt}\frac{\partial U}{\partial v_x}$$

in comparison with equation (8.66), where the generalised potential energy is

$$U = q\Phi - q\mathbf{A}\cdot\mathbf{v} \tag{8.67}$$

With this notation, the Lagrangian for a charged particle moving in an electromagnetic field is given by

$$L = T - U = T - q\Phi + q\mathbf{A}\cdot\mathbf{v} \tag{8.68}$$

This was the Lagrangian used earlier in article 8.6 to derive the generalised momentum of a particle in the electromagnetic field.

8.9 RAYLEIGH'S DISSIPATION FUNCTION

Another point regarding Lagrange's equation must be noted. Only if some of the forces acting on the system are derivable from a potential, Lagrange's equations assume the form

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_j} - \frac{\partial L}{\partial q_j} = Q_j \tag{8.69}$$

where L contains only those forces that are conservative while Q_j includes the forces that are not derivable from a potential. An illustration of this latter type of force is the frictional force. Many times we come across a situation when the frictional force is proportional to velocity. The x-component of this force may be written as

$$F_{fx} = -k_x v_x \tag{8.70}$$

where k_x is the x-component of the frictional force per unit velocity in the x-direction. Forces of this type are derivable from Rayleigh's dissipation function $\mathcal{F}$ defined by

$$\mathcal{F} = \tfrac{1}{2}\sum_i (k_x v_{i_x}^2 + k_y v_{i_y}^2 + k_z v_{i_z}^2) \tag{8.71}$$

where $i = 1, 2, \ldots, n$ covers all the particles of the system.

It is obvious that

$$F_f = -\nabla_v \mathcal{F} \tag{8.72}$$

where we have introduced a new symbol

$$\nabla_v = \mathbf{i}\frac{\partial}{\partial v_x} + \mathbf{j}\frac{\partial}{\partial v_y} + \mathbf{k}\frac{\partial}{\partial v_z}$$

which is the vector velocity differential operator.

To explain the physical significance of Rayleigh's dissipation function, let us calculate the work done by the system against friction as

$$dW_f = -\mathbf{F}_f\cdot d\mathbf{r} = -\mathbf{F}_f\cdot\mathbf{v}\,dt$$

$$= (k_x v_x^2 + k_y v_y^2 + k_z v_z^2)\,dt$$

Therefore,

$$\frac{dW_f}{dt} = (k_x v_x^2 + k_y v_y^2 + k_z v_z^2) = 2\mathcal{F}$$

Thus, the rate of dissipation of energy by friction is equal to twice Rayleigh's dissipation function.

Component Q_j of the generalised force arising as a result of frictional force is given by

$$Q_j = \sum_i \mathbf{F}_{fi} \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} = -\sum \nabla_v \mathcal{F} \cdot \frac{\partial \mathbf{r}_i}{\partial q_j}$$

$$= -\sum \nabla_v \mathcal{F} \cdot \frac{\partial \dot{\mathbf{r}}_i}{\partial \dot{q}_j}, \text{ by equation (8.25)}$$

$$= -\frac{\partial \mathcal{F}}{\partial \dot{q}_j}$$

Substituting this value of Q_j, we can write Lagrange's equations as

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right) - \frac{\partial L}{\partial q_j} + \frac{\partial \mathcal{F}}{\partial \dot{q}_j} = 0 \tag{8.73}$$

Thus, if frictional forces are acting on the system, we must specify two scalar functions—the Lagrangian L and Rayleigh's dissipation function $\mathcal{F}$—to derive the equations of motion.

QUESTIONS

1. What are constraints? Explain, giving examples, the meaning of holonomic and nonholonomic constraints.
2. Explain the meaning of scleronomous and rheonomous constraints. Give illustrations of each.
3. Is the Lagrangian formulation more advantageous than the Newtonian formulation? Why?
4. What do you understand by cyclic coordinates? Show that the generalised momentum corresponding to a cyclic coordinate is a constant of motion.
5. Explain the term 'virtual displacement' and state the principle of virtual work.
6. When is a generalised force the same as the corresponding component of conventional force? When is it different?
7. Describe the use of Rayleigh's dissipation function.
8. What arbitrariness exists in the definition of the Lagrangian?
9. How is a generalised potential defined? What advantage does it have over the conventional potential?
10. What do you understand by generalised momentum? Express the momentum conservation theorem in terms of generalised momenta.
11. State D'Alembert's principle in words.
12. Define the Hamiltonian. When is it equal to the total energy of the system? When is it conserved?
13. What is meant by velocity-dependent potential? Where do we come across such a potential?
14. What is meant by a configuration space? How is this concept used

to describe the motion of a system of particles?

15. Generalised coordinates may not have the dimensions of distance, generalised force may not have the dimensions of force, still their product has the dimensions of work. Explain.
16. While solving a problem in mechanics, a lot of labour is saved by choosing the appropriate coordinate system. The choice is usually guided by the symmetry (spherical, cylindrical, etc.) possessed by the system. In the following cases find out the most convenient coordinate system (try cartesian, spherical polar and cylindrical systems). Further find out the degrees of freedom and all the equations of constraint.
    (i) a double pendulum
    (ii) a disc rolling down an inclined plane
    (iii) a particle constrained to move on the surface of a sphere
    (iv) a particle moving along the surface of a right circular cone
    (v) a missile on a submarine
    (vi) a dumb bell whose centre is constrained to move along a circle
17. Is it possible to express all the *fundamental* forces in nature in the form of generalised potentials? Explain.
18. What is meant by a non-conservative force? Give some examples of such forces. Is the Lorentz force non-conservative? Explain.
19. Conditions for the conservation of generalised momenta corresponding to cyclic coordinates are more general than the two momentum conservation theorems. Comment.

PROBLEMS

1. Write down the convenient generalised coordinates for: (a) a missile in a submarine, free to point in any direction, (b) a point on the wheel of the bicycle, (c) a double pendulum, (d) a spherical pendulum. Also write down the equations of constraints.
2. Write down the Lagrangian in terms of convenient coordinates for: (a) a particle of mass m and charge q, in a uniform gravitational field and the electric field of an infinite vertical line charge of density σ, (b) a planet in the field of the sun and rotating about a fixed axis, (c) two particles of masses m_1 and m_2 and charges q and $-q$ respectively, in a uniform gravitational field. Also obtain the equations of constraints in each case.
3. A cylinder, initially at rest, rolls down an inclined plane without slipping. Investigate its motion using Lagrange's equations. What ensures rolling without slipping? Can we still use Lagrange's equations? Explain.

4. A Kater's pendulum has a radius of gyration k and mass M and its centre of mass is at distance L below the point of support. A pin is attached to it at distance d above the support and from this hangs a simple pendulum of mass m and length l. Obtain the equations of motion of the simple pendulum. Solve these for $M \gg m$. Explain why we get the unphysical result that the simple pendulum has infinite amplitude at resonance.
5. A spring of force constant k is confined to move along a vertical line with its upper end fixed. A simple pendulum of length l and mass m is suspended from its lower end. Obtain the Lagrangian equations and show that they are equivalent to Newton's equations.
6. Show, from Lagrange's equations, that the orbit of a particle in a central force field lies in a plane.
7. Two particles of masses m_1 and m_2 lie on a smooth table connected by a spring of unstretched length l, force constant k and mass m. Obtain the constants of motion and frequency of vibration in the absence of rotation.
8. A simple pendulum of mass m has support of mass M. The support is free to slide on a horizontal frictionless plane. Obtain and solve the equations of motion.
9. A double pendulum vibrates in a vertical plane. If masses of the bobs are m_1 and m_2 and the lengths of the pendulums are l_1 and l_2 respectively, obtain the Lagrangian and Lagrange's equations for this double pendulum.
10. Two particles of masses m_1 and m_2 are connected by a string which passes through a hole in a smooth table so that m_1 rests on the table surface and m_2 hangs underneath. Assuming that m_2 moves only in a vertical line, find the suitable generalised coordinates for the system. Write down Lagrange's equations for the system and discuss their physical significance, if any. Reduce the problem to a single second-order differential equation and obtain a first integral of the equation. What is its physical significance? (Consider the motion only so long as neither m_1 nor m_2 passes through the hole.)
11. A hoop rolls without slipping along a horizontal circle and is free to tilt. Obtain its equations of motion. Derive the condition for rolling at constant inclination. Find the horizontal and vertical components of the force at the point of contact.
12. A body is sliding down an inclined plane which is moving horizontally with constant velocity. Find the position of the body as the function of time.
13. Show that for a hoop on a plane, the constraint of rolling without slipping is not holonomic. Write down the constraint in differential form.
14. Show that the kinetic energy of a hoop rolling without slipping on a

horizontal surface is

$$ma^2(\dot{\psi} + \dot{\varphi} \cos \theta)^2 - \frac{3}{4} ma^2\dot{\theta}^2 - \frac{ma^2}{4} \sin^2 \theta \, \dot{\varphi}^2$$

Include the effect of gravity and find equations for θ, φ and ψ.

15. Masses m and $2m$ are suspended from the ends of a string of length l_1 which passes over a pulley. Masses $3m$ and $5m$ are similarly suspended from the ends of a string of length l_2 which passes over another pulley. These two pulleys hang from the ends of a string of length l_3 which passes over a fixed pulley. Set up Lagrange's equations and find the acceleration and the tension in the strings.

16. Consider a system in which mass m is suspended by a weightless rigid rod from the end of a spring which is constrained to move in a vertical line. Obtain the equation of motion of the system.

17. A bead is able to slide along a smooth wire in the shape of parabola $z = cr^2$. The parabola is rotating with constant angular velocity ω about its vertical symmetry-axis. Write down the equation of motion of the bead.

18. Consider a double Atwood's machine. In this, mass M_1 is connected to one end of a string which passes over a pulley. From the other end is suspended another pulley round which passes another string. From the ends of this string are suspended masses M_2 and M_3. Assume that the masses of the pulleys and friction are negligible. Select a proper set of coordinates and obtain Lagrange's equations for the system. Solve these for the acceleration of masses.

19. Consider a simple, plane pendulum which consists of mass m attached to a string of length l. After the pendulum is set into motion, the length of the pendulum is shortened at a constant rate,

$$\frac{dl}{dt} = \text{const}$$

The point of suspension remains fixed. Compute the Lagrangian and Hamiltonian functions. Compare the Hamiltonian and the total energy and discuss the conservation of energy of the system.

20. A body subject to gravity moves on a smooth sphere along a vertical great circle. Find its angular acceleration assuming that the great circle is at rest in an inertial frame.

21. Set up Lagrange's equations in polar coordinates and use them to find the force law for a particle moving along the orbit

$$\frac{1}{r} = ae^{b\varphi}$$

in an inertial frame with radius vector $\mathbf{r}$ sweeping out the area at a constant rate.

22. Use Lagrange's equation to find an expression for the angular acceleration of a pendulum bob moving in a vertical plane. Also find the tension in the string which supports the bob.

23. Viscous force F retarding a sphere in a fluid of coefficient of viscosity η is given by Stokes's law,

$$F = -6\pi\eta a v$$

where v is the terminal velocity of the sphere. Set up the corresponding dissipation function. Then set up Lagrange's equation for the sphere falling vertically under the action of gravity in the fluid. Solve it for terminal velocity.

24. An inductor and a capacitor are connected in parallel in a circuit containing an alternating e.m.f. Obtain the Lagrangian function for this system.

25. A particle of mass m moves without friction on a straight wire inclined at angle θ with the vertical z-axis. If the wire rotates about the z-axis with constant angular velocity ω, find the Lagrange's equation of motion and its solution.